

MUHAMMAD ABU BAKAR

AI / ML Engineer

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Location: Lahore, Pakistan

PROFILE SUMMARY

AI Engineer with 2+ years of experience building production-grade Agentic AI systems, RAG Pipelines, Voice-to-Voice AI Agents, and LLM-powered automation workflows. Specialized in designing intelligent, scalable systems using LangChain, LangGraph, VAPI, n8n, FastAPI, and Supabase. Proven ability to deliver real-world AI solutions that automate complex business processes, reduce manual effort, and enhance operational efficiency across diverse industries. Passionate about bridging cutting-edge AI research with real-world product development from ideation to full production deployment.

TECHNICAL SKILLS

Languages:	Python (Proficient), JavaScript, C/C++ (Basic)
Artificial Intelligence:	Machine Learning, Deep Learning, NLP, LLM, Agentic AI, RAG Pipelines, Generative AI, AI Automation, Voice-to-Voice AI Agents
Frameworks & Libs:	LangChain, LangGraph, LangSmith, CrewAI, FastAPI, Flask, React, Next.js, NumPy, Pandas, Scikit-learn, TensorFlow/Keras, OpenCV
Databases & Vector:	Supabase, PostgreSQL, Pinecone, NeonDB, Faiss, Chroma
Tools & Platforms:	n8n, Make.com, VAPI, Retell AI, ElevenLabs, Hugging Face, Postman, Twilio, AWS
Deployment & VC:	Vercel, Netlify, Railway, Render, Docker, Git, GitHub
Software:	VS Code, Jupyter Notebook, Google Colab, Kaggle
Soft Skills:	Adaptability, Time Management, Quick Learning, Effective Multitasking

PROFESSIONAL WORK EXPERIENCE

AI Developer — Spadasoft	Lahore, Pakistan Oct 2025 – Present
- Designed and deployed Agentic AI pipelines, RAG systems, AI chatbots, and n8n automation workflows for client projects, reducing manual operations and enabling intelligent business automation. - Built Voice-to-Voice AI Agents with STT/TTS integration, an AI-powered Appointment Booking Chatbot, and a Client Project Monitoring System as part of the company's internal product development.	

AI Developer — FlintSol

Lahore, Pakistan | Mar 2025 – Oct 2025

- Designed and implemented LLM/GPT-based RAG systems and chatbots to enhance language understanding, automate workflows, and support customer knowledge retrieval.
- Developed AI-driven automation solutions, including social media auto-posting and n8n workflows, reducing manual effort and improving operational efficiency.

AI Developer — Software Alliance

Lahore, Pakistan | Aug 2024 – Apr 2025

- Designed and implemented LLM- and GPT-based RAG systems to enhance language understanding and automate complex workflows, and developed machine learning models to improve prediction accuracy and support data-driven decision-making.

CERTIFICATIONS

Cisco Networking Academy — Programming Essentials in Python	Aug 2024
Great Learning — Introduction to Natural Language Processing	Sep 2024
Innovista — Machine Learning	Sep 2024
CampusX — LangChain & LangGraph	July 2025
Microsoft Azure — Fundamentals of AI and ML	Sep 2025
Udemy — AI Agents & Agentic AI	Apr 2026

PROJECTS

WAWCD (WhatsApp Web Chrome Dimension) Product

OpenAI, AssemblyAI, Whisper AI, Google Calendar API, WhatsApp API, Shopify, Chrome Extension

- Implemented Voice-to-Voice Agent with STT/TTS pipeline, RAG-based Knowledge Base with PDF extraction, and AI Guardrails for response safety.
- Integrated Appointment Booking Chatbot with Google Calendar sync, automated WhatsApp follow-ups, and Shopify connectivity for e-commerce management.

ID Bridge — AI Governance & Monitoring Platform Product

Next.js, React, Firebase, Firestore, Chrome Extension, TypeScript, Vercel AI

- Built a Chrome Extension tracking employee AI tool usage across platforms, sending privacy-safe metadata to a Firebase backend via secure APIs.
- Developed an admin dashboard with usage analytics, user activity views, and CSV/PDF export.

AI-Driven Inbound Call Management System (Voice-to-Voice AI Agent)

OpenAI, Retell AI, Twilio, n8n, Next.js

- Engineered a production Voice-to-Voice AI system automating real-time inbound client calls with natural conversational responses, Twilio telephony integration, and n8n workflows handling intelligent call routing, CRM updates, and automated follow-ups.

AI-Powered Social Media Automation System

n8n, OpenAI, Pexels, Captions.ai, Google Drive, Meta APIs

- Built an end-to-end AI automation pipeline generating content, video descriptions, and captions using OpenAI, storing assets in Google Drive, and auto-publishing across TikTok, Facebook, LinkedIn, and YouTube.

AI-Powered WhatsApp Automation & Appointment Booking Agent

n8n, Twilio, WhatsApp API, Google Calendar, OpenAI

- Built an Agentic AI system automating WhatsApp conversations with intelligent auto-replies, handling incoming queries in real-time, and seamlessly booking appointments directly into Google Calendar.

AI-Powered Multiple Pages Facebook Post Scheduler System (56+ Pages)

Meta API, Flask, NeonDB, HTML/CSS

- Built an AI-powered scheduling system automating publishing of posts, reels, and videos across 56+ Facebook pages within minutes, with intelligent commenting workflows to boost engagement at scale.

RAG-Based Sports Court Booking Chatbot

OpenAI, LangChain, Pinecone, GraphQL, React

- Built a RAG-powered chatbot for a sports club enabling real-time court availability checks, time slot management, and seamless booking functionality using Pinecone vector search, OpenAI, and GraphQL APIs.

AI-Powered Price Prediction & Validation Telegram Bot

Python, Flask, n8n, Machine Learning, Telegram Bot API

- Built an ML-powered Telegram chatbot predicting and validating client-specific pricing with a real-time feedback loop, enabling users to approve or adjust predictions directly via chat for continuous model improvement.

AI-Powered Data Entry Automation - OCR System

Python, AWS Textract, Pandas, OpenPyXL, Flask

- Built an OCR-based data entry automation system extracting structured data from PDF documents using AWS Textract, automatically converting and organizing output into Excel reports eliminating manual data entry effort entirely.

EDUCATION

Bachelor of Science in Data Science — [Virtual University](#)

Lahore, Pakistan | 2024 – 2028